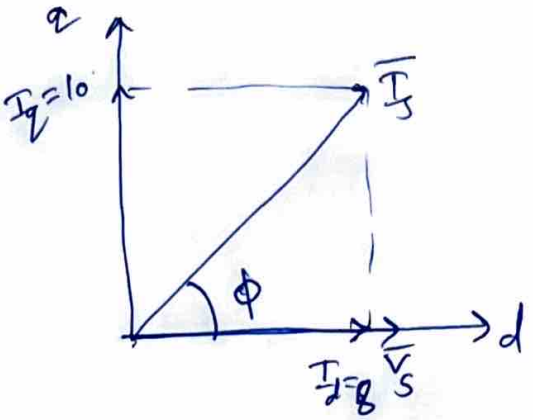
Assignment – 6

Q.2.

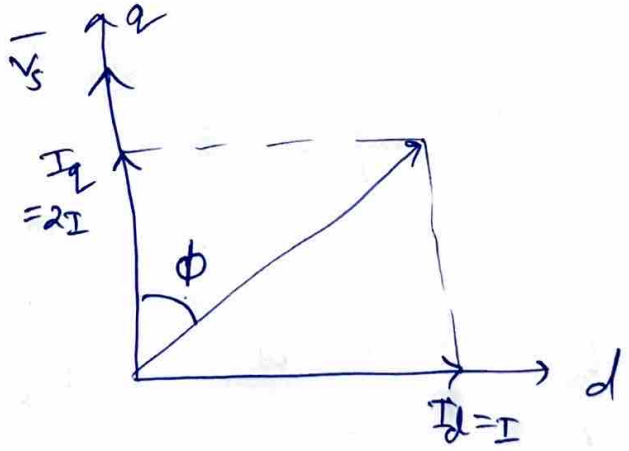


$$\tan\phi = \frac{I_q}{I_d} = \frac{10}{8}.$$

$$\Rightarrow \phi = 51.34^\circ$$

$$\Rightarrow \boxed{\cos\phi = 0.6246}$$

Q.3.



$$\Rightarrow \tan\phi = \frac{I}{2I} = 1/2$$

$$\Rightarrow \phi = 26.56^\circ$$

$$\Rightarrow \boxed{\cos\phi = 0.894}$$

Q.6.

$$(V_{DS,1})_{max} = V_{in}$$

Q.8.

$$t_{on,S1} = 10\,\mu s$$

$$t_{off,S1} = 40\,\mu s$$

$$\Rightarrow D = \frac{10\,\mu s}{10\,\mu s + 40\,\mu s} = \frac{1}{5}$$

$$\Rightarrow V_0 = n \cdot D \cdot V_{in} \qquad \left(n = \frac{N_S}{N_P}\right)$$

$$= \frac{2}{5} \cdot \frac{1}{5} \cdot 400$$

$$V_0 = 32$$

Q.9.

$$\tau_S = \frac{L}{R} = \frac{12\,mH}{10\,m\Omega} = 1.2$$

Now, $\tau_S = \frac{K_{PC}}{K_{IC}} \Rightarrow \frac{K_{PC}}{K_{IC}} = 1.2$